Quantifying the Impact of Lift-off on the Optic Nerve Tract of Space Crew

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Abstract

In this note we demonstrate the impact of severe vibrations in a passenger vehicle upon a nerve tract's conduction of action potentials, under the ephaptic geometric setting first considered in 2016-2017. As an application, this work could be used to determine how nerve signals of crew optic tracts might be scrambled during the various phases of space travel, from the point of view of space vehicle vibrations.

The lift-off phase of launching a rocket has a high vibrational impact on the rocket's structure which in turn must have some impact on the crew vehicle and the crew (for example, [1]). In order to quantify the significance of this impact for *neural information processing*, we modified the MATLAB program developed in [2, 3] by allowing the inter-axonal distance matrix (denoted in [2] as W_{ij}) to be perturbed by noise in each entry, even as time evolved. When the added noise variance in each entry was 0.01 in magnitude, we obtained baseline-like propagation (i.e., similar to what is obtained in the absence of noise). This is shown in Figure 1. When the added noise variance was augmented by another additive term of 0.01 magnitude, so that the total variance was 0.02, double the previous value, the profile of propagation was significantly altered. Please see Figure 2. The action potential initiation time on axon 3 was severely delayed as compared to baseline, so that the third axon fired *after* the second axon, instead of before it. This demonstrates a clear impact on time-coded information processing due to an axon tract being immersed in a field of mechanical vibrations. In the case of an optic nerve carrying visual information to the brain, we can surmise that these signals would now carry altered informational content to subsequent processing centers.

References

[1] Curtis E. Larsen. Nasa experience with pogo in human spaceflight vehicles. In *Proceedings of the NATO RTO Symposium ATV-152 on Limit-Cycle Oscillations and Other Amplitude-Limited, Self-Excited Vibrations*, Norway, May 2008. NATO Research and Technology Organization, NASA Langley

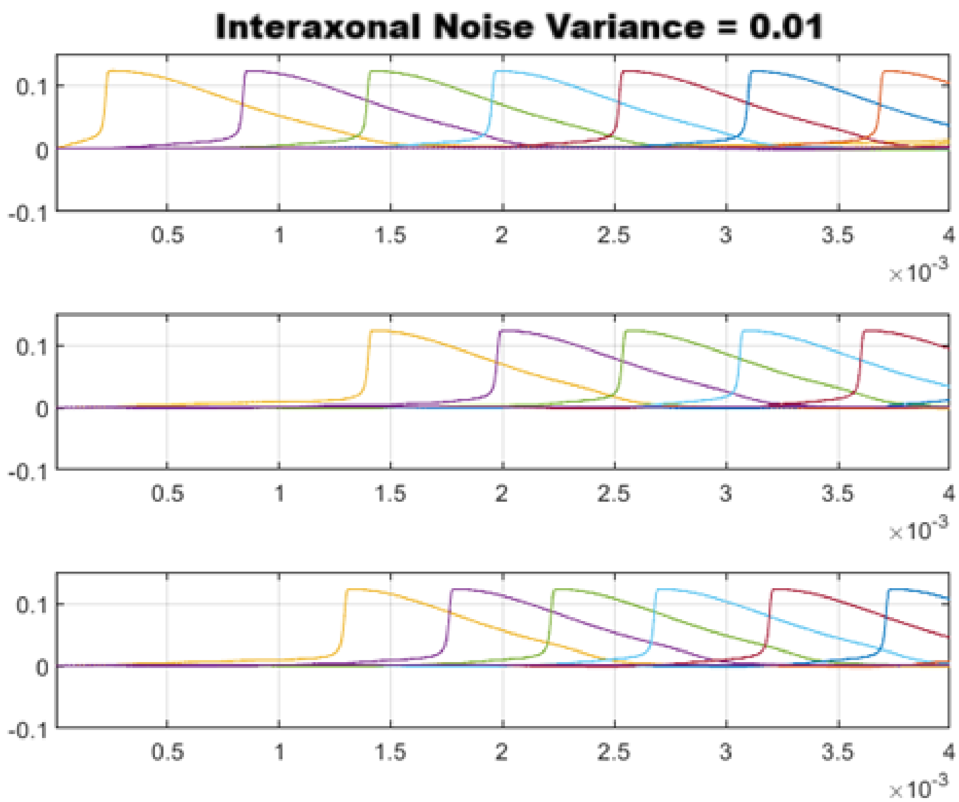

Figure 1: Low noise case, with geometric noise variance = 0.01. The axons are numbered from one to three going from top subplot to bottom subplot. In each subplot, the horizontal axis is time in seconds and the vertical axis is voltage in Volts. Differently colored temporal voltage profiles in a subplot represent distinct nodes firing one after another in saltatory propagation.

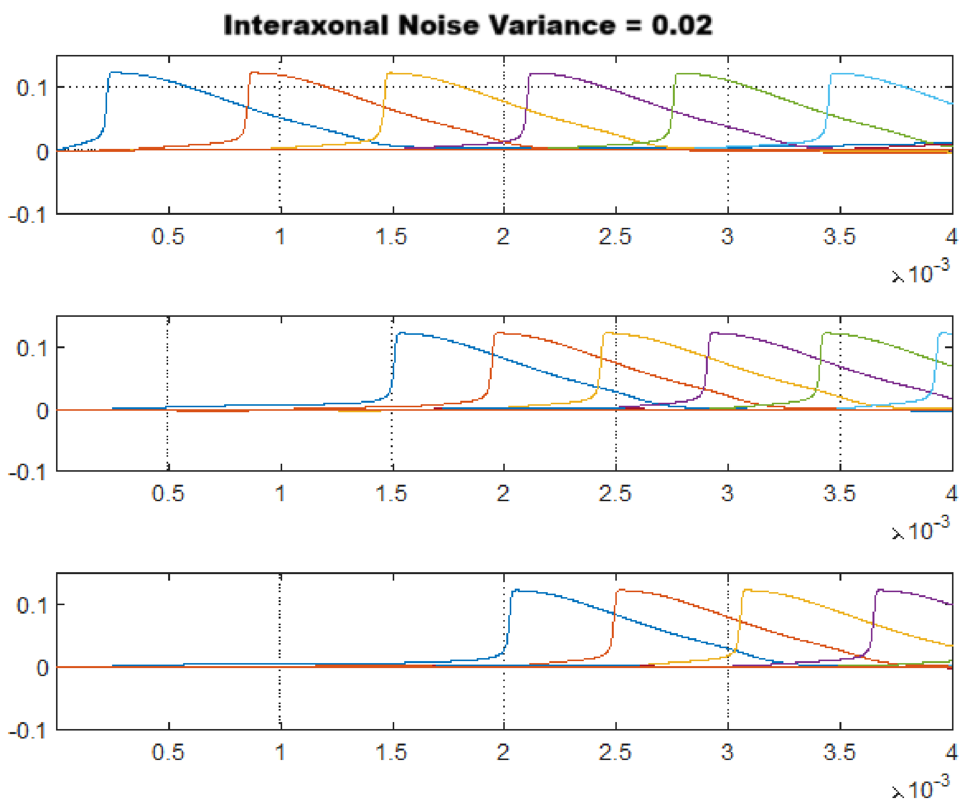

Figure 2: High noise case, with geometric noise variance = 0.02. The axons are numbered from one to three going from top subplot to bottom subplot. In each subplot, the horizontal axis is time in seconds and the vertical axis is voltage in Volts. Differently colored temporal voltage profiles in a subplot represent distinct nodes firing one after another in saltatory propagation.

Research Center. NASA Document ID: 20080018689, Report Number: RTO-MP-AVT-152.

[2] Aman Chawla, Salvatore Morgera, and Arthur Snider. On axon interaction and its role in neurological networks. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2019.

[3] Aman Chawla. *On Axon-Axon Interaction via Currents and Fields*. PhD thesis, University of South Florida, 2017.